

Quiz 12. Φ Age student
20 yes

Yes

$$P(\text{age} \leq 30 | \text{yes}) = 2/7$$

$$P(\text{student} = \text{yes} | \text{yes}) = 6/7$$

No

$$P(\text{age} \leq 30 | \text{no}) = 3/5$$

$$P(\text{student} = \text{yes} | \text{no}) = \frac{1}{5}$$

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

$$P(y | \hat{x}) = C \hat{x}^{(buy = yes)} P(buy = yes)$$

$$= \frac{2}{7} \times \frac{6}{7} \times \frac{9}{14}$$

$$= 0.095$$

$$P(N | \hat{x}) = P(\hat{x} | (buy = no)) P(buy = no)$$

$$= \frac{3}{5} \times \frac{1}{5} \times \frac{5}{14}$$

$$= 0.043$$

$\therefore \Phi$ age 20 is student no yes